Uncertainty-Aware Semantic Split Computing

Solving the energy-accuracy tradeoff in continuous remote seizure monitoring through closed-loop adaptive sensing.

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The Inherent Limits of Continuous Monitoring

The Extremes

Full Edge Approach

Raw EEG → Wearable Device → Deep CNN (L1-L5) → Prediction

Structural Flaws

- Deep CNNs too heavy for microcontrollers
- Severe memory & thermal constraints
- Low accuracy due to limited model complexity

Full Cloud Approach

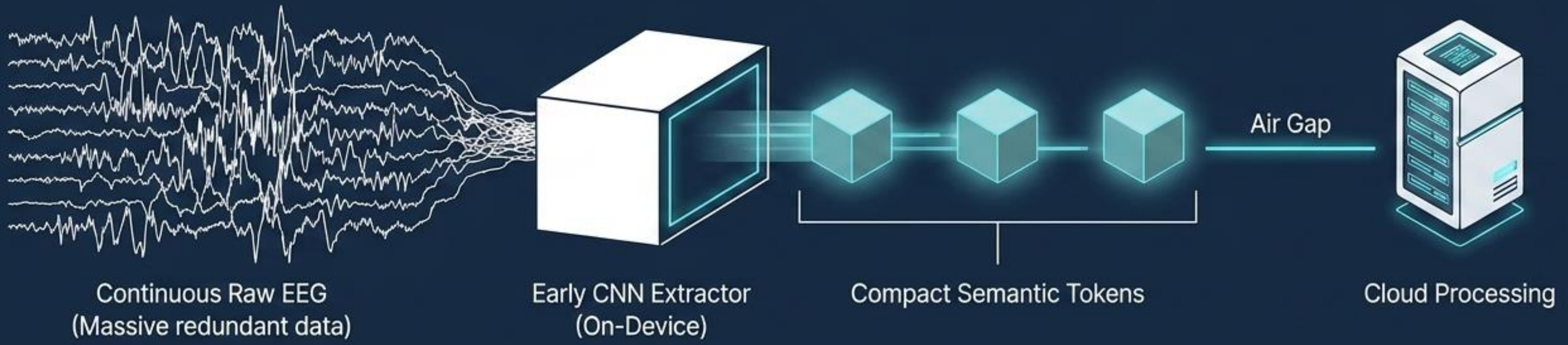
Raw EEG → Transmit Full Raw Data → Cloud Server → Deep CNN (L1-L5) → Prediction

Structural Flaws

- Massive continuous bandwidth consumption
- High communication energy drains batteries
- Vulnerable to network latency

Key Problem: Current systems optimize for accuracy, not energy.
Devices lack the intelligence to know *when* to sense intensely, making communication the biggest energy consumer.

Transmitting Meaning, Not Raw Waveforms



The Shift

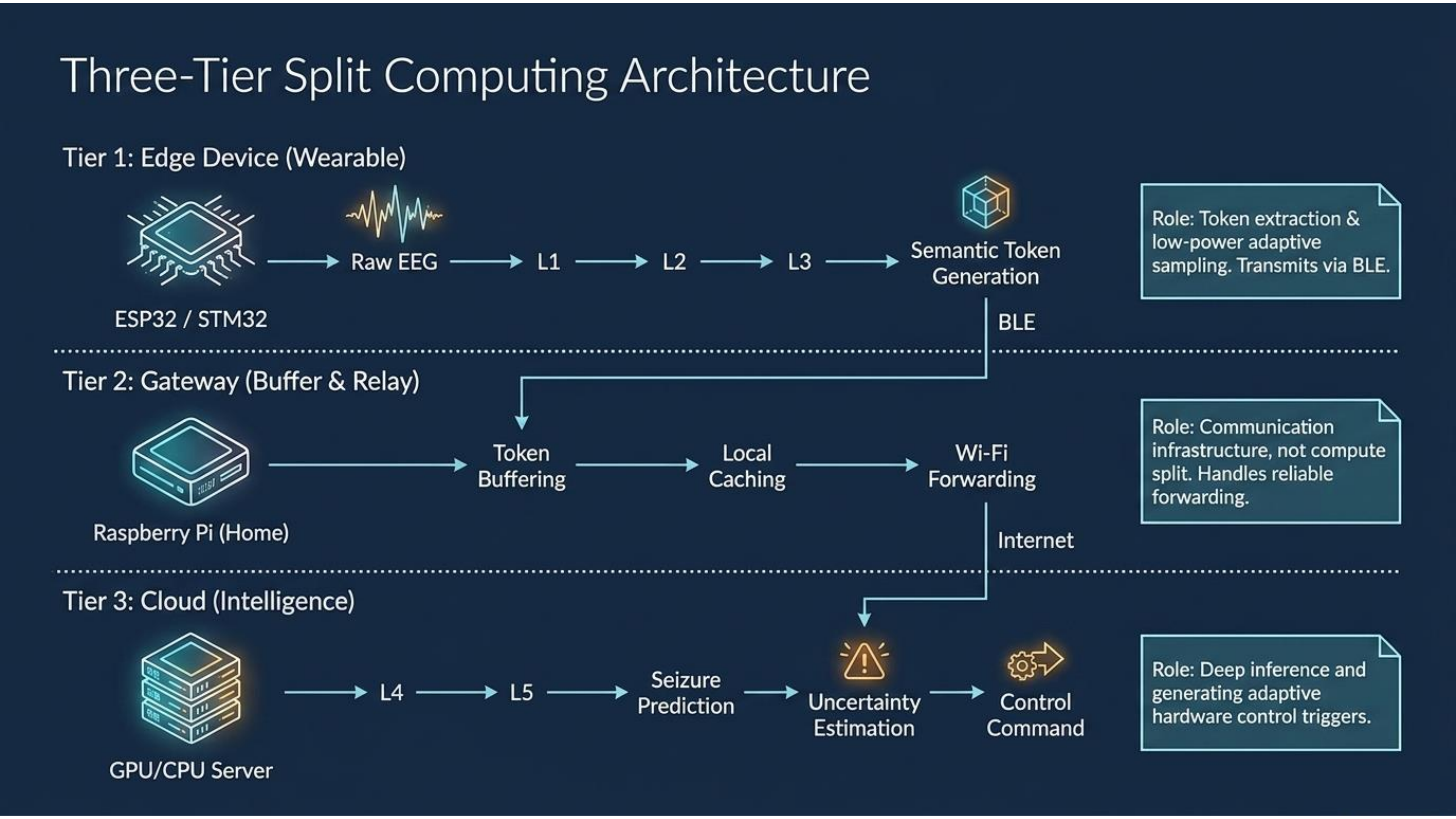
Replace continuous fixed-rate raw signal transmission with discrete semantic token communication.

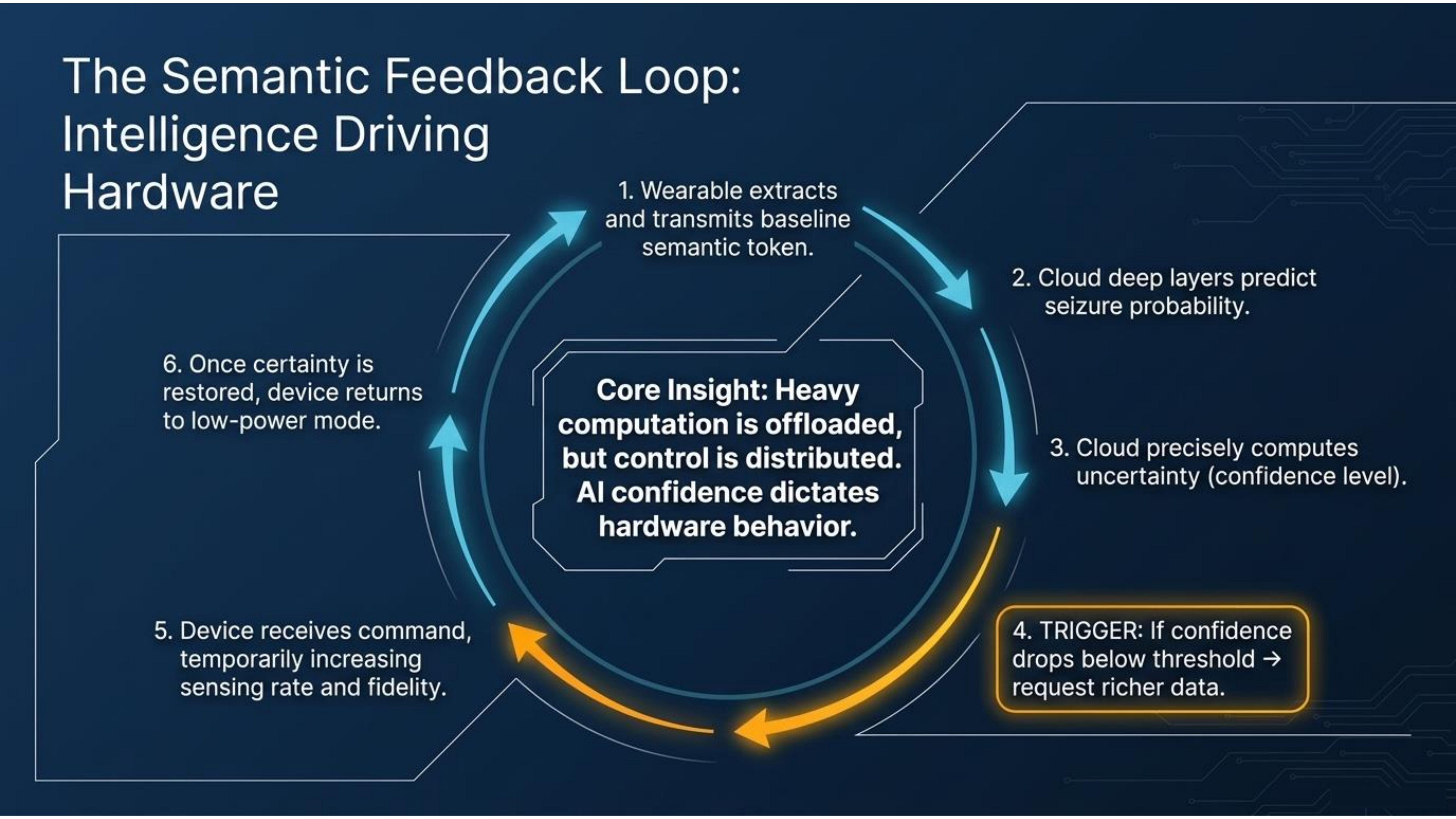
The Mechanism

Early CNN layers on the wearable extract vital features locally, sending only meaningful representations over the air.

The Result

Drastically reduced packet size, lowered energy consumption, and a foundation for adaptive sensing.


Three-Tier Split Computing Architecture
Tier 1: Edge Device (Wearable)
ESP32 / STM32
Raw EEG
L1
L2
L3
Semantic Token Generation
Role: Token extraction & low-power adaptive sampling. Transmits via BLE.
BLE
Tier 2: Gateway (Buffer & Relay)
Raspberry Pi (Home)
Token Buffering
Local Caching
Wi-Fi Forwarding
Role: Communication infrastructure, not compute split. Handles reliable forwarding.
Internet
Tier 3: Cloud (Intelligence)
GPU/CPU Server
L4
L5
Seizure Prediction
Uncertainty Estimation
Control Command
Role: Deep inference and generating adaptive hardware control triggers.


The Semantic Feedback Loop: Intelligence Driving Hardware
1. Wearable extracts and transmits baseline semantic token.
2. Cloud deep layers predict seizure probability.
3. Cloud precisely computes uncertainty (confidence level).
4. TRIGGER: If confidence drops below threshold → request richer data.
5. Device receives command, temporarily increasing sensing rate and fidelity.
6. Once certainty is restored, device returns to low-power mode.
Core Insight: Heavy computation is offloaded, but control is distributed. AI confidence dictates hardware behavior.

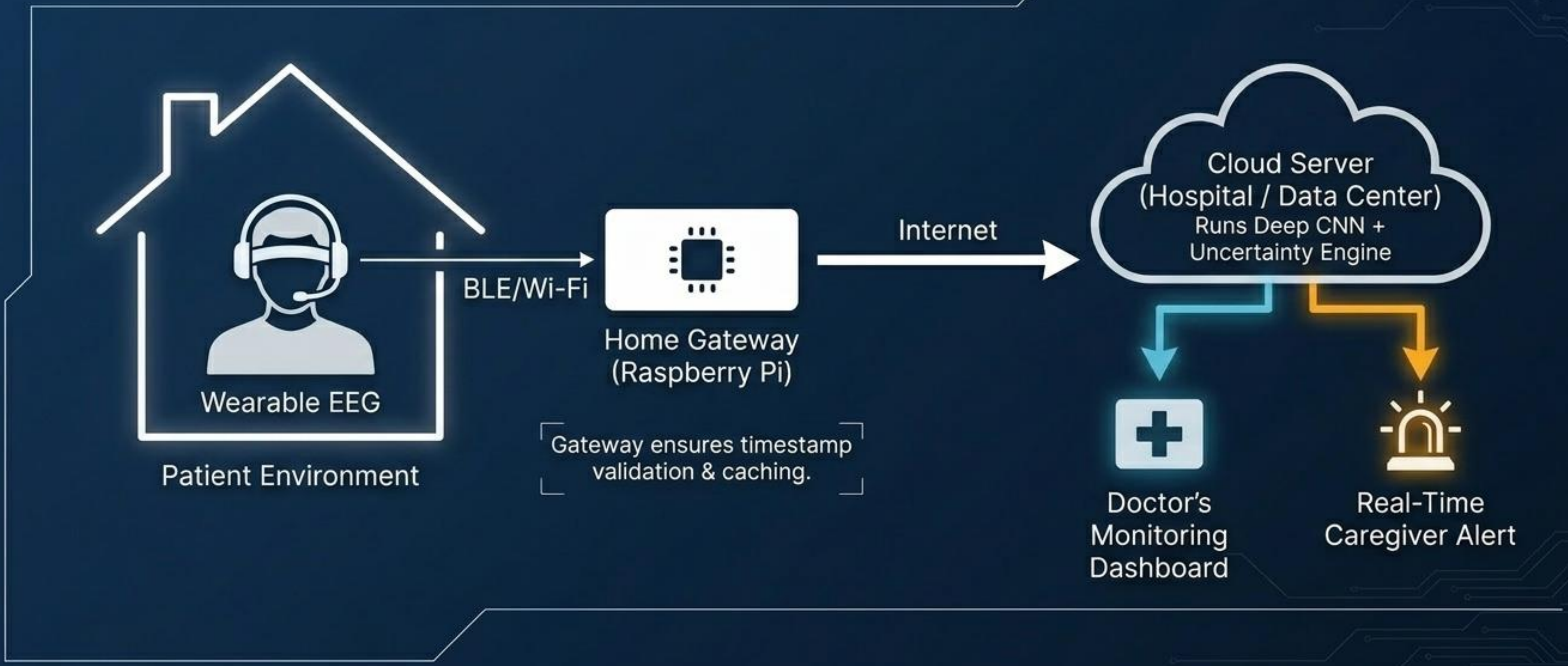
Paradigm Shift: A Dimension-by-Dimension Comparison

Paradigm Comparison Matrix

Feature	Traditional Systems	Our System
Data Sent	Raw EEG Waveforms	Compressed Semantic Tokens
Sampling	Fixed	Adaptive (Demand-driven)
Intelligence Architecture	Static (All-Edge or All-Cloud)	Uncertainty-Aware Split Compute
Battery Life	Low (Drained by transmission)	Highly Optimized
Feedback Loop	None (Open-loop sensing)	Yes (Closed control loop)

Visual Synthesis: Individually, CNNs, split inference, and feature compression are not novel. The breakthrough is the combination of uncertainty-driven sensing closing the control loop in biomedical IoT.

Deployment Scenario: Hospital-Level Monitoring at Home



System Validation Framework: The Bonn EEG Dataset

Dataset Profile

- Public benchmark by Andrzejak et al. (2001)
- 173.61 Hz sampling rate
- Single-channel EEG
- 23.6-second segments (100 per subset)
- Lightweight footprint enables fast CNN prototyping before clinical scaling.

Dataset Diagnostic Table

Set Z	Surface EEG	Healthy subjects	Eyes open
Set O	Surface EEG	Healthy subjects	Eyes closed
Set N	Intracranial EEG	Epileptic patients	Seizure-free (hippocampus)
Set F	Intracranial EEG	Epileptic patients	Seizure-free (epileptogenic zone)
Set S	Intracranial EEG	Epileptic patients	Active seizure

Implementation Reality: Friction at the Edge

Proposed Architecture		Practical Implementation (Reality)
Compute Constraints	CNN encoder deployed directly on Raspberry Pi gateway.	⚠ Pi faced severe power limits/reboots and PyTorch compatibility issues. **Pivot:** Executed on a laptop-based edge gateway for stable PyTorch execution.
Data Acquisition	Real-time stream from physical wearable sensors.	⚠ Hardware unavailable. **Pivot:** Laptop simulated sensor streaming, injecting the Bonn dataset into the pipeline.
Infrastructure Wins	Reliable edge-to-cloud token passing.	✔ Successful. Serial comms (ESP32/Pi) validated, and laptop successfully forwarded semantic tokens to AWS cloud server via the Pi gateway.

Crucial Insight: Model splitting works conceptually, but edge hardware limits dictate physical layout during early prototyping.

Bridging the Gap: From Dataset to Clinical Reality

Multi-Modal Sensor Fusion

Single-channel EEG is insufficient for deployment.

Integrating ECG (cardiac correlation) and Accelerometers (motion artifacts) is required to drastically reduce false positives.

Stable Closed-Loop Control

Simple threshold switching creates volatile systems.

Devices require hysteresis and state-based control to prevent mode oscillations and unnecessary energy spikes.

Communication-Aware Inference

Real networks suffer packet loss. The system must evolve to feature adaptive transmission (prioritizing critical tokens) and local fallback inference when disconnected.

The Horizon: Scaling the Smart Medical IoT Ecosystem

Hardware Evolution

Hardware Evolution

Upgrade edge AI using Raspberry Pi 5, Jetson Nano, or Coral TPUs to resolve deep learning bottlenecks. Implement dedicated powered hubs.

Algorithmic Refinement

Algorithmic Refinement

Optimize semantic token size via aggressive quantization and neural compression. Transition to patient-specific models to handle long-term signal drift.

Broad Clinical Impact

Expand beyond seizures to continuous cardiac monitoring. Enable true telemedicine and rural healthcare by reducing hospital infrastructure burden without sacrificing diagnostic fidelity.